

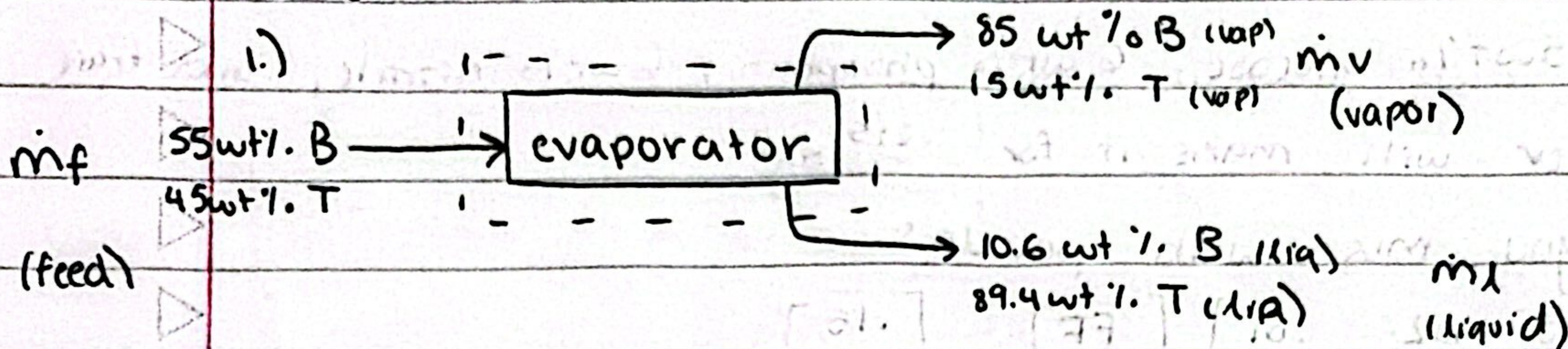
2.) 2.39

3.) 2.58

4.) 2.6

5.) 2.64

Homework H3



$$b) \dot{m}_f = \dot{m}_v + \dot{m}_l$$

$$B: .55 \dot{m}_f = .85 \dot{m}_v + .106 \dot{m}_l$$

$$T: .45 \dot{m}_f = .15 \dot{m}_v + .894 \dot{m}_l$$

$$.55(\dot{m}_v + \dot{m}_l) = .85 \dot{m}_v + .106 \dot{m}_l$$

$$.55 \dot{m}_v + .55 \dot{m}_l = .85 \dot{m}_v + .106 \dot{m}_l$$

$$.444 \dot{m}_l = .3 \dot{m}_v$$

$$\frac{\dot{m}_v}{\dot{m}_l} = .676$$

c.) dof

4a - 82

4b - 0

5a - 0

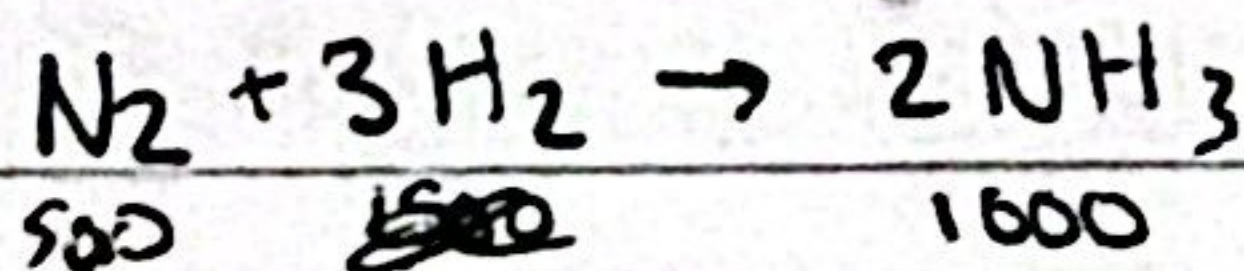
5b - 0

5c - 0

5d - 2

2 - 2 = 0 $\Rightarrow$ specified.

2.) N:H is 1:3

25% of $N_2 \rightarrow NH_3$, producing 1000 lbmol/h of NH_3 

$$\frac{500 \text{ lbmol/h}}{\text{total}} = .25 \text{ total} \Rightarrow 2000 \text{ lbmol/h of } N_2$$

if 2000 lbmol/h of N_2 , then 6000 lbmol/h of H_2

total is 8000 lbmol/h

$$PV = nRT$$

$$V = \frac{nRT}{P}$$

$$R = 35.9 \frac{\text{ft}^3 \cdot \text{atm}}{\text{lbmol} \cdot ^\circ R}$$

$$R = .7302 \frac{\text{ft}^3 \cdot \text{atm}}{\text{lbmol} \cdot ^\circ R}$$

$$100 \cdot V = 8000 \cdot .7302 \cdot 1440$$

$$V = 84119.04 \text{ ft}^3/\text{h}$$

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13/15

2 pts

20/20

\$ 2.58

4 2.60

5 2.64

3) 15wt% glucose, 6wt% phosphate, 6wt% nitrate, and trace
 Supplier will make it for \$15/kg
 or you mix with powders.

$$\begin{bmatrix} .45 & .02 & .01 \\ 0 & .35 & .15 \\ 0 & .12 & .58 \end{bmatrix} \begin{bmatrix} FF \\ SG \\ FN \end{bmatrix} = \begin{bmatrix} .15 \\ .06 \\ .06 \end{bmatrix}$$

G P N

$$\begin{bmatrix} FF \\ SG \\ FN \end{bmatrix} = \begin{bmatrix} .325 \\ .139 \\ .075 \end{bmatrix}$$

$$\begin{aligned} & .325(1.45g + .02p + .01n) + .139(6g + .35p + .15n) \\ & + .075(6g + .12p + .58n) \\ & = .4625g + .0065p + .00325n + .04865p + .02085n \\ & + .009p + .0435n \\ & = .4625g \end{aligned}$$

a) make a mixture of 32.5% Fast-feed, 13.9% super-gro,
 7.5% formula N, and 46.1% water.

To make 1 kg of broth, use 325g of f-f, 139g of
 s-g, and 75g of f-n.

b.) cost

$$.325(20)x + .139(10)x + .075(15)x = 9.015x$$

homemade cost \$9.02/kg, while supplier cost \$15.00/kg.

You will save \$5.98/kg

How of batches?
 - 5 pks

$$c.) .325(.08) + .139(.08) = .02087 = \boxed{2.09 \text{ wt\% trace}}$$

$$\begin{bmatrix} .48 & .02 & .01 & .03 \\ 0 & .35 & .15 & .08 \\ 0 & .12 & .58 & 6 \end{bmatrix} \begin{bmatrix} FF \\ SG \\ FN \end{bmatrix}$$

set basis

$$p=1$$

$$.45g + .02p + .01n + .03t =$$

$$.35p + .15n + .08t =$$

$$.12p + .58n =$$

$$.45g = .15$$

$$g=3$$

$$.02p + .35p + .12p = .06$$

$$.02p + .35p = .06$$

$$.01n + .15n + .58n = .06$$

$$.03t + .08t = .03$$

with 1

$$.45x_1 = .15 \quad .02x_1 + .35x_2 + .12x_3 = .06$$

$$.01x_1 + .15x_2 + .58x_3 = .06 \quad .03x_1 + .08x_2 = .03$$

$$\text{set } x_1 = 3$$

$$.35x_2 + .12x_3 = 0$$

$$.15x_2 + .58x_3 = .03$$

$$.08x_2 = -.06$$

$$.58x_3 = .1425$$

$$x_2 = -.75$$

$$x_3 = .25$$

it's not possible because x_2 is a negative number.

(not possible to remove Super-Gro)

25/30

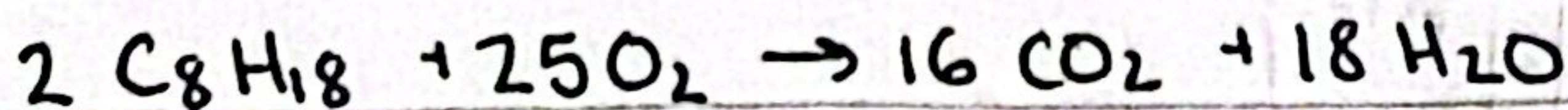
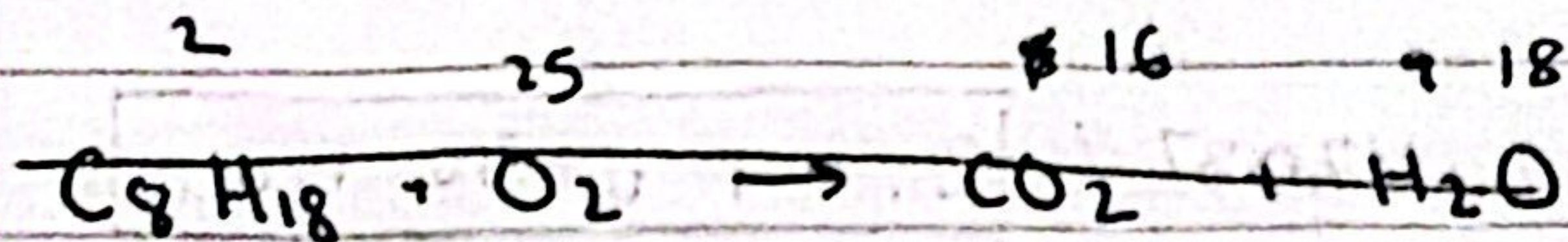
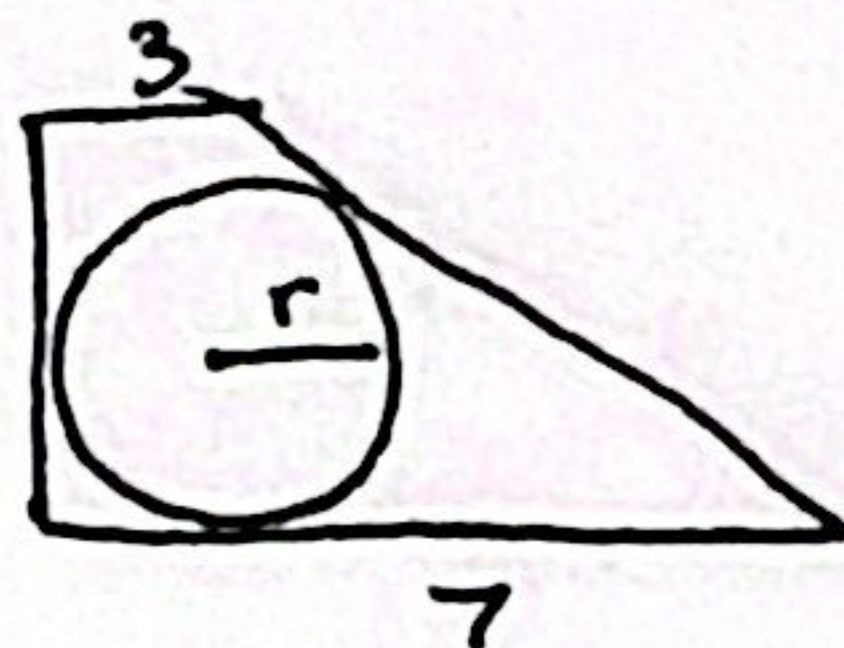
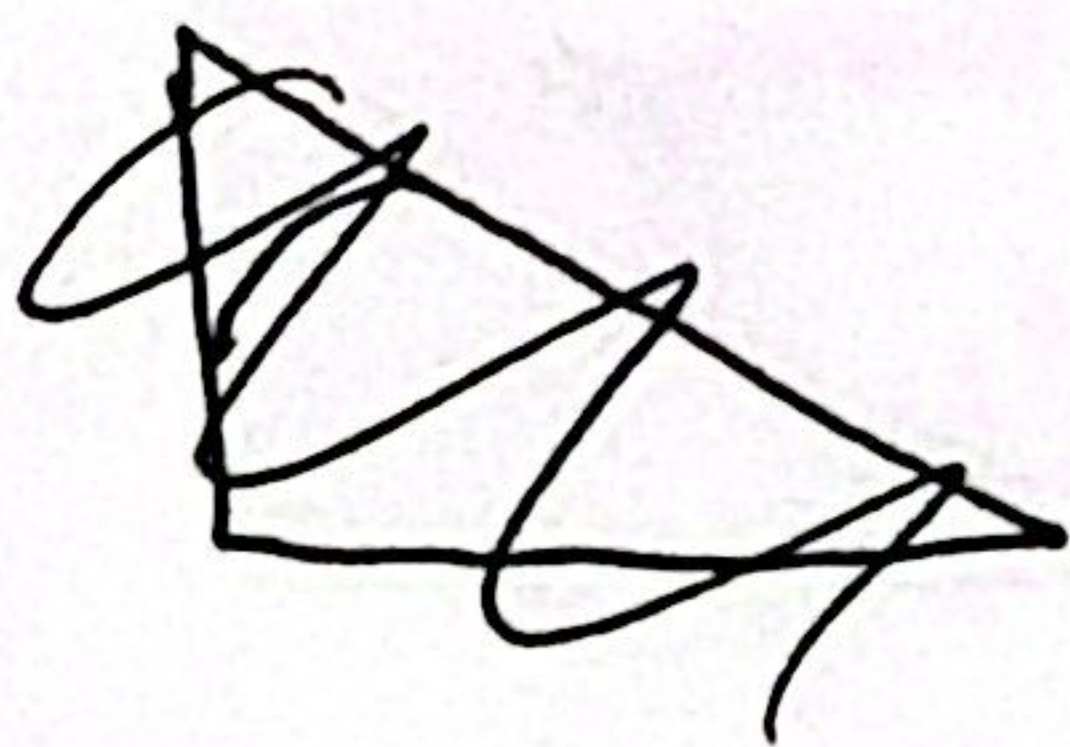
4.) 5.6 bil tons of CO_2 , 300 mil people, lbs/person?

$$\rightarrow \frac{5600 \text{ mil tons} \times 2000 \text{ lbs}}{1 \text{ ton}} = 11200000 \text{ mil lbs}$$

$$\frac{11200000 \text{ mil lbs}}{300 \text{ mil ppl}} = \boxed{37333.33 \text{ lbs/person}}$$

seems extreme, but take into account the extremely wealthy ppl

15/15



$$\rho_g = \frac{.7}{\rho_{\text{water}}} = \frac{.7}{1} \quad \rho_{\text{gas}} = .7 \text{ g/mL} = 700 \text{ g/L} \quad 1 \text{ gal} = 3.785 \text{ L}$$

$$\frac{700 \text{ g}}{1 \text{ L}} \times \frac{3.785 \text{ L}}{1 \text{ gal}} = 2649.5 \text{ g/gal} \times 27.5 \text{ mi/gal} = 96.35 \text{ g/mi}$$

the avg person drives ~14000 mi a year

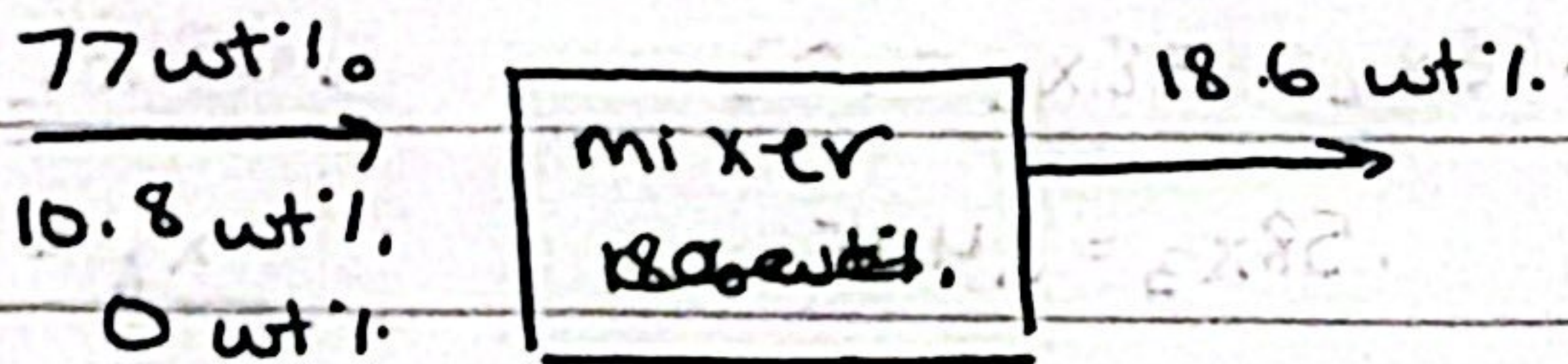
$$\frac{1348900 \text{ g}}{\text{person}} \times \frac{1 \text{ lb}}{453.6 \text{ g}} = 2973.76 \text{ lbs/person}$$

→ very similar value

5.) 200 kg/day, contains 18.6 wt% H_2SO_4

only given 77% wt% H_2SO_4 , and 10.8 wt% H_2SO_4 , & water.

conc is 10 g/kg, dilu is 2 g/kg



4a.) 2

4b.) 1

5a.) 0

5b.) 1

5c.) 0

5d.) 1

$$5 - (1 + 1) = 3$$

completely specified

$$3 - 2 = 1$$

(not fully specified)

$$(a \cdot .77) + (b \cdot .108) = (.186 \cdot 200) \text{ minimize: } 10a + 2b = \text{Cost}$$

$$77a + 10.8b = 37.2$$

$$a + b + c = 200$$

$$(.186 \cdot 200) - .77a = .108b$$

$$c = 200 - a - b$$

$$b = \frac{37.2 - .77a}{.108}$$

$$10.8b = .186 \cdot 200 - .77a$$

$$10a + \frac{37.2 - .77a}{.108} \cdot 2 = \text{Cost}$$

$$C = 200 - a - \frac{37.2 - .77a}{.108}$$

$$\text{Cost} = 688.89 - 4.26a$$

$$5.13a - 144.44$$

basis, let $c=0$ (aka no water)

$$c = 5.13a - 144.44 = 0$$

$$a = 28.16 \text{ kg/day}$$

~~minimizes~~ minimizes this cost which is lowk most expensive

$$200 - (143.67 + 28.16) = C$$

$$b = \frac{37.2 - .77a}{.108} = \frac{37.2 - .77(28.16)}{.108}$$

$$b = 143.67 \text{ kg/day}$$

$$C = 28.17 \text{ kg/day}$$

17/20